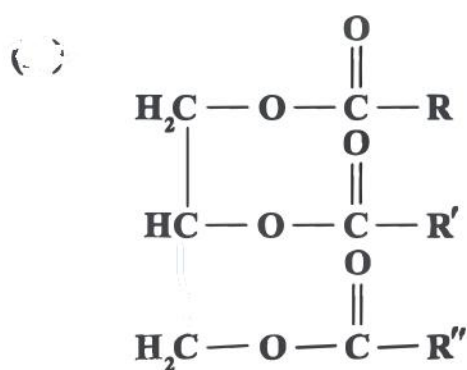
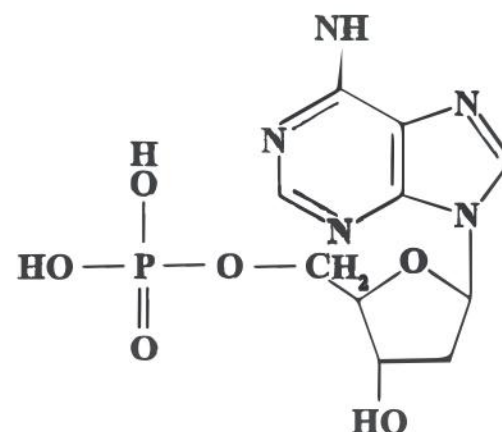


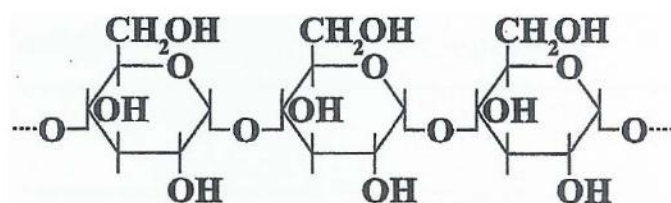
1. Which of the following molecules BEST represents the structure of a triglyceride?



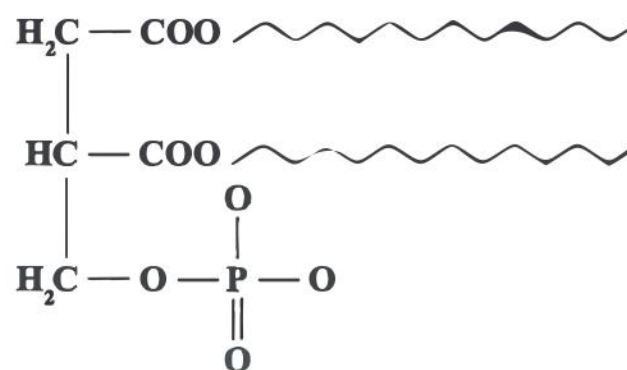
(C)



(B)



(D)



2. Which of the following is NOT a polymer?

- (A) Glycogen
(B) Cellulose
(C) Amylose
(D) Glucose

4.

- Which of the following cell structures contain phospholipids?

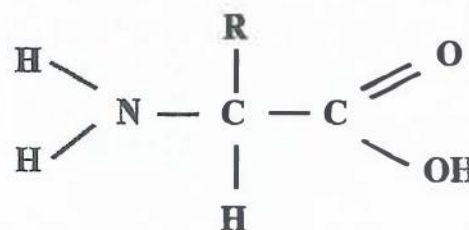
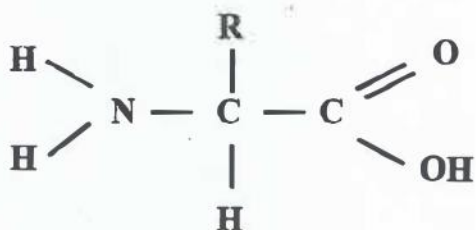
- I. Ribosome
II. Mitochondrion
III. Endoplasmic reticulum

3. A diet high in triglycerides may lead to obesity. The feature of triglycerides which MOST likely contributes to this is that they

- (A) are insoluble in water
(B) have many C-H bonds
(C) contain glycerol molecules
(D) have many -OH groups

- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

Item 5 refers to the following molecules.



5. Which of the following bonds is formed between these two molecules during a condensation reaction?
- (A) Ester
(B) Peptide
(C) Hydrogen
(D) Glycosidic
-
6. Which of the following structures is found in BOTH a prokaryotic cell and an eukaryotic cell?
- (A) Ribosome
(B) Chloroplast
(C) Mitochondrion
(D) Nuclear membrane
7. Collagen exhibits a secondary level of protein structure and organization because
- (A) it contains four polypeptide chains
(B) each chain forms a helix held by covalent bonds
(C) each chain forms a helix held by hydrogen bonds
(D) it is made up of repeated units of three different amino acids
8. Which of the following reactions is needed for the formation of a peptide bond?
- (A) Oxidation
(B) Reduction
(C) Hydration
(D) Condensation
9. When a plant cell is placed in a solution with a less negative water potential than its cell sap, the cell becomes more
- (A) turgid because water diffuses into the cell
(B) flaccid because water diffuses into the cell
(C) turgid because water diffuses out of the cell
(D) flaccid because water diffuses out of the cell

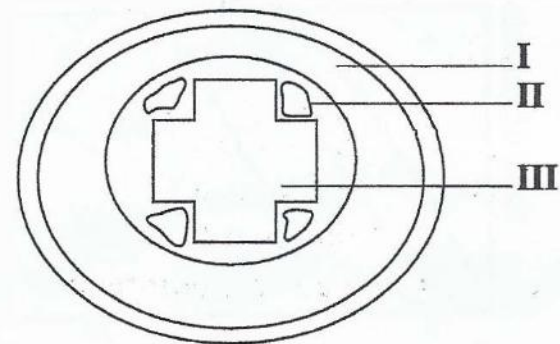
10. Which of the following combinations forms evidence which BEST supports the endosymbiont theory?

	Prokaryotic Cells	Eukaryotic Cells
(A)	Contain circular DNA in cells	Contain linear DNA in mitochondria
(B)	Contain circular DNA in cells	Contain circular DNA in mitochondria
(C)	Contain circular DNA in mitochondria	Contain linear DNA in mitochondria
(D)	Contain linear DNA in mitochondria	Contain circular DNA in mitochondria

11. The table below shows three processes involved in the movement of substances across cell membranes. Which of the following correctly states whether the three processes shown are active or passive in nature?

	Diffusion	Endocytosis	Osmosis
(A)	Active	Active	Active
(B)	Active	Passive	Active
(C)	Passive	Active	Passive
(D)	Passive	Passive	Passive

- Item 12 refers to the following diagram which shows a cross section of the root of a dicotyledon.



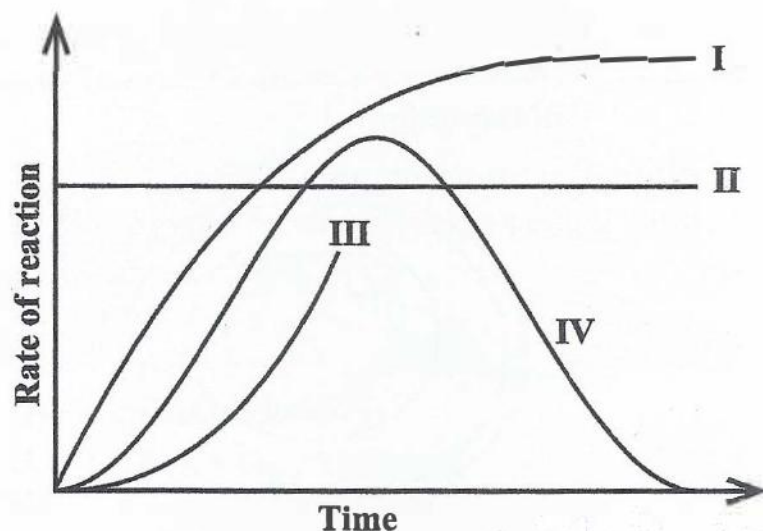
12. Which of the following correctly identifies the structures labelled I, II and III?

	I	II	III
(A)	Cortex	Phloem	Xylem
(B)	Cortex	Xylem	Phloem
(C)	Phloem	Cortex	Xylem
(D)	Xylem	Phloem	Cortex

13. Which of the following is the BEST example of anabolism?

- (A) Glycogen is converted to glucose.
- (B) Peptide bonds are hydrolysed by peptidases.
- (C) Amino acids are joined by peptide bonds to form a polypeptide.
- (D) Amylose and amylopectin combine to form starch molecules.

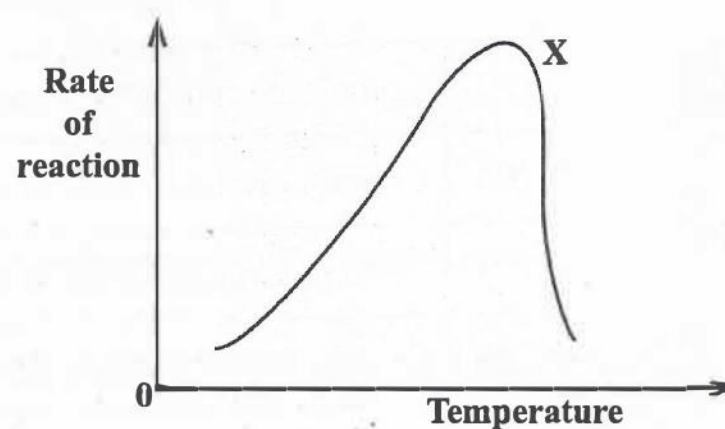
Item 14 refers to the following activity curves which show the effects of different conditions on the activity of enzymes.



14. Which of the activity curves BEST represents the rate of reaction of an enzyme with increasing substrate concentration?

- (A) I
- (B) II
- (C) III
- (D) IV

Item 15 refers to the following graph which shows the progress of an enzyme-catalysed reaction with increasing temperature.



15. The decline in the rate of reaction at X is MOST likely the result of

- (A) peptide bonds being broken
- (B) substrate concentration becoming too low
- (C) the enzyme losing its tertiary structure
- (D) the enzyme losing its secondary structure

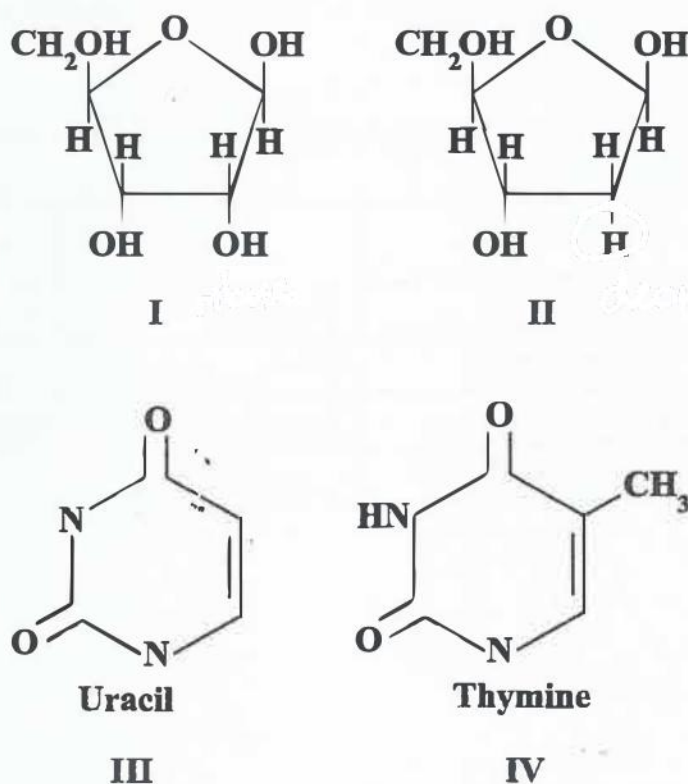
Item 16 refers to the following events which take place during the semi-conservative replication of DNA in a eukaryotic cell.

- I. Covalent bonds form between adjacent nucleotides on the same strand
- II. The DNA double helix unwinds
- III. Free nucleotides join by hydrogen bonds to those on the exposed strand
- IV. Hydrogen bonds are broken between the complementary base pairs

16. Which of the following correctly identifies the sequence of events, from first to last, which takes place during semi-conservative replication of DNA in a eukaryotic cell?

	FIRST → LAST			
(A)	I	II	III	IV
(B)	II	III	IV	I
(C)	IV	II	I	III
(D)	II	IV	III	I

Item 17 refers to the following diagrams of molecules which are involved in the formation of nucleic acids.



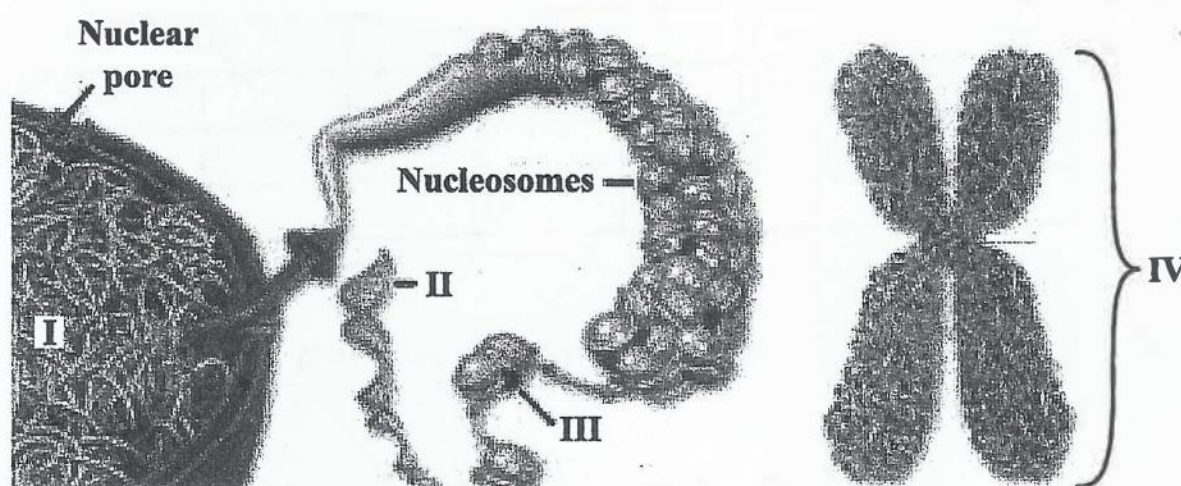
17. Which TWO molecules when combined with a phosphate group form an RNA nucleotide?

- (A) I and III
- (B) I and IV
- (C) II and III
- (D) II and IV

18. Which of the following events in meiosis contributes directly to variation?

- (A) Formation of spindle fibres
- (B) Separation of homologous pairs
- (C) Attachment of bivalents to spindle fibres
- (D) Genetic rearrangement between non-sister chromatids

Item 19 refers to the following diagram (not drawn to scale) which shows nuclear material in a cell.



19. Which of the following rows matches each structure with its correct name?

	I	II	III	IV
(A)	Chromosome	DNA	Histone	Chromatin
(B)	Chromosome	Chromatin	DNA	Chromatid
(C)	Chromatin	DNA	Histone	Chromatid
(D)	DNA	Histone	Chromatid	Chromosome

Item 20 refers to the following information.

The mRNA genetic code for six amino acids are shown below. The RNA triplet code UAG acts as a stop codon.

mRNA Genetic Code	Amino Acid
GUU	Valine
UUC	Phenylalanine
UAC	Cysteine
AAA	Lysine
AGU	Serine
AUG	Methionine

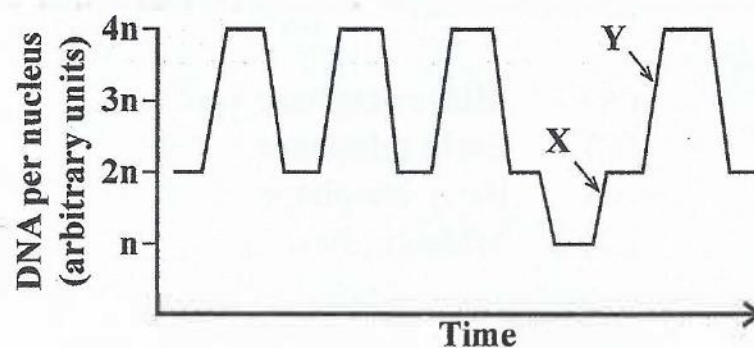
20. Which of the following is the correct amino acid sequence for the following DNA template strand, TAC AAG ATG TCA ATC?

- (A) Methionine – phenylalanine – valine – serine
 (B) Methionine – phenylalanine – cysteine – serine
 (C) Methionine – valine – serine – lysine – phenylalanine
 (D) Methionine – valine – cysteine – phenylalanine – lysine

21. Which of the following statements is NOT true of most homologous chromosomes?

- (A) They are the same length.
 (B) They contain the same alleles.
 (C) They contain the same number of genes.
 (D) They contain the same heterochromatin bands.

Item 22 refers to the following diagram which represents the relative amounts of DNA per nucleus during many cell divisions in an animal cell.



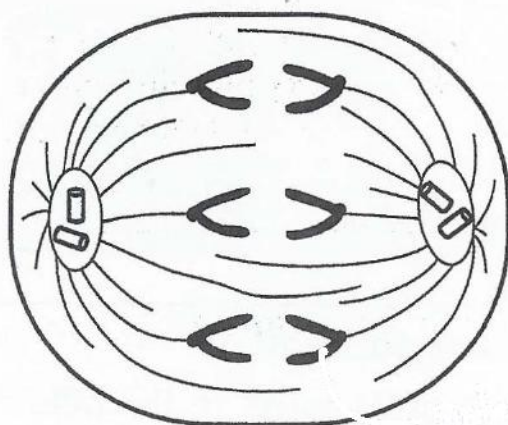
22. Which of the following processes are occurring at X and Y?

	X	Y
(A)	Fertilization	DNA synthesis
(B)	DNA synthesis	Separation of chromosomes
(C)	DNA synthesis	Fertilization
(D)	Fertilization	Cytokinesis

23. In populations of Pacific salmon, most fish are either very large or very small. This is because sperm from medium-sized fish seldom fertilize the eggs. The allele frequency of medium-sized fish is therefore very low. The allele frequency of medium-sized fish is MOST likely

(A) artificial selection
☐ (B) disruptive selection
☐ (C) stabilizing selection
☐ (D) directional selection

Item 24 refers to the following diagram showing one stage of mitotic cell division.



24. From the position and shape of the chromosomes, what stage of **mitosis** has been reached?

(A) Mid-metaphase
☐ (B) Early telophase
☐ (C) Early anaphase
☐ (D) Mid-prophase

25. There are various isolating mechanisms which can result in populations of the same species evolving into two distinct species. Which of the following is an example of behavioural isolation?

(A) *Pinus radiata* flowers in February while *Pinus attenuata* flowers in April.
☐ (B) *Viola arvensis* grows on calcareous soils whereas *Viola tricolor* prefers acidic soils.
☐ (C) The flower structure of certain orchids prevents the transfer of pollen from another species.
☐ (D) In species of funnel web spiders, only males which spray the female with a chemical substance are allowed to mate with her.

26. In sickle-cell anaemia, the sickling of red blood cells results from the mutation of

(A) a single base pair in the gene that encodes for the alpha-globin polypeptide
☐ (B) a single base pair in the gene that encodes for the beta-globin polypeptide
☐ (C) two base pairs in the gene that encodes for the alpha-globin polypeptide
☐ (D) two base pairs in the gene that encodes for the beta-globin polypeptide

27. Which of the following events that occur during meiosis is MOST likely to result in chromosomal mutation?

☐ (A) Non-disjunction
☐ (B) DNA replication
☐ (C) Homologous pairing
☐ (D) Independent assortment

Item 28 refers to the following information.

Fruit colour in summer squash is controlled by a pair of genes occupying a different locus from the gene that controls colour. "W" represents absence of colour (white) and "w" represents presence of colour in the fruit. The allele for yellow fruit is "Y" and for green fruit is "y".

28. What is the phenotypic ratio of the F_2 from a cross between two F_1 with the genotype $WwYy$?

- (A) 9 yellow : 3 green : 4 white
- (B) 9 yellow : 4 green : 3 white
- (C) 12 white : 1 yellow : 3 green
- () 12 white : 3 yellow : 1 green

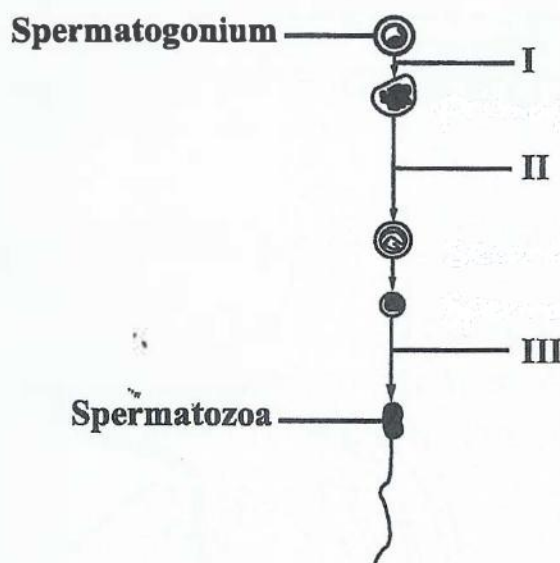
29. In addition to the human insulin gene, which of the following enzymes would be required to produce a recombinant plasmid?

- (A) ADPase
- () DNA ligase
- (C) RNA polymerase
- (D) Reverse transcriptase

30. One role of transfer RNA in protein synthesis is to

- () deliver amino acids to ribosomes
- (B) terminate the elongation of the polypeptide chain
- (C) act as a supporting framework for polypeptide chains
- (D) carry protein-building instructions from DNA into the cytoplasm

Item 31 refers to the following diagram which shows the stages of spermatogenesis in a mammal.



31. Which of the following accurately lists the type of cell division occurring at I, II and III?

	I	II	III
(A)	Mitosis	Mitosis	Meiosis I
()	Mitosis	Meiosis I	Meiosis II
(C)	Meiosis I	Mitosis	Meiosis II
(D)	Mitosis	Mitosis	Meiosis II

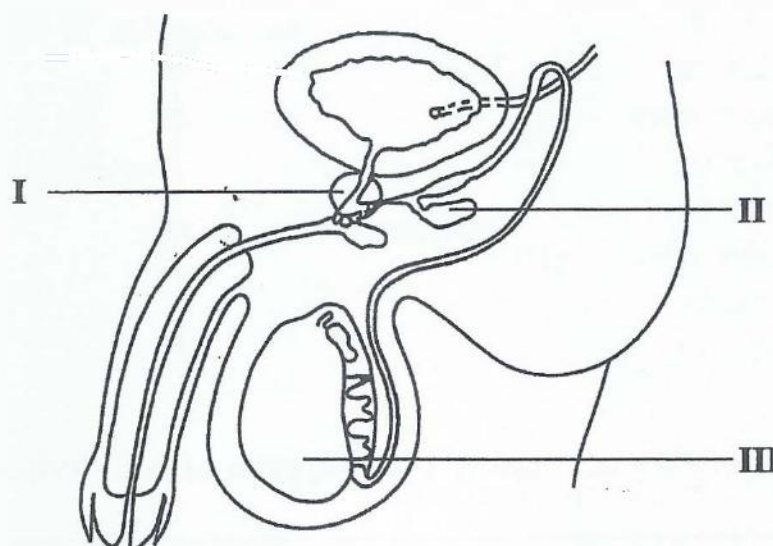
32. The MOST important advantage of **sexual** reproduction is that

- (A) it eliminates the risk of deleterious mutations
- (B) many offspring are produced in a short period of time
- (C) it allows adaptation to changing environmental conditions
- (D) desirable genes are always passed on to the next generation

33. Which of the following accurately describes the outbreeding mechanisms in flowering plants used to promote cross-pollination?

	Outbreeding Mechanism		
	Self-incompatibility	Protandry	Dioecious
(A)	Fertilization may occur	Stigmas mature before stamens	Male and female flowers on separate plants
(B)	No fertilization	Stamens mature before stigmas	Male and female flowers on separate plants
(C)	Fertilization may occur	Stamens mature before stigmas	Male and female flowers on same plant
(D)	No fertilization	Stigmas mature before stamens	Male and female flowers on same plant

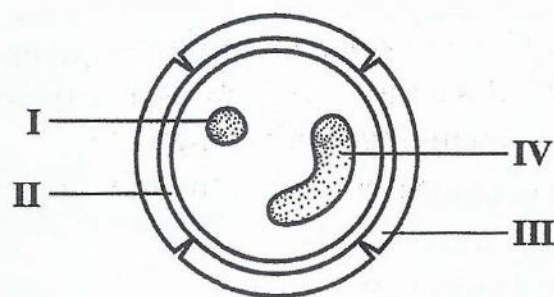
Item 34 refers to the following diagram of the human male reproductive system as seen in lateral view.



34. Which of the labelled structures produce fluids which aid in the transport of sperm?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Item **35** refers to the following diagram showing a mature pollen grain.



35. Which of the labelled structures will give rise to two male gametes?

- (A) I
- (B) II
- (C) III
- (D) IV

36. Which row in the following table matches each structure with its correct number of chromosomes?

	Chromosome Number		
	n	2n	3n
(A)	Pollen tube nucleus	Cotyledon	Endosperm
(B)	Pollen tube nucleus	Endosperm	Cotyledon
(C)	Cotyledon	Pollen tube nucleus	Endosperm
(D)	Cotyledon	Endosperm	Pollen tube nucleus

37. Which of the following events occur during double fertilization in plants?

- I. A male gamete fuses with the egg cell in the embryo sac to form a triploid nucleus.
 - II. A male gamete fuses with the egg cell in the embryo sac to form a diploid zygote.
 - III. A male gamete fuses with the diploid nucleus in the embryo sac to form a triploid nucleus.
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III

38. Which of the following statements identifies a difference between the secondary oocyte and the ovum?

- (A) The secondary oocyte is diploid while the ovum is haploid.
- (B) The secondary oocyte has two polar bodies while the ovum has one.
- (C) The secondary oocyte is surrounded by the zona pellucida while the ovum is not.
- () The secondary oocyte is at Metaphase II of meiosis while the ovum has completed the meiotic divisions.

39. Which of the following explains why the nucleus of the ovum occupies a much smaller portion of the cell than does the nucleus in a sperm?

- (A) The sperm nucleus is swollen.
- () The ovum contains food stores in the cytoplasm.
- (C) The chromosomes in the ovum are more tightly coiled.
- (D) The nucleus of the ovum has fewer chromosomes than the sperm nucleus.

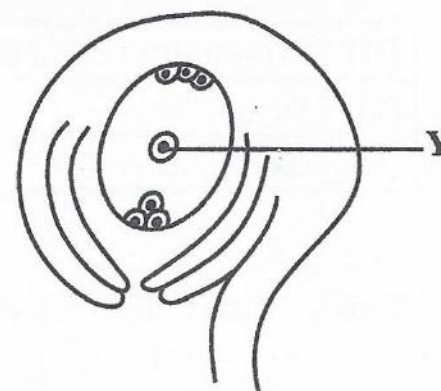
40. Which of the following pairs of structures contributes directly to the development of the placenta?

- (A) Yolk sac and allantois
- (B) Chorion and allantois
- () Chorion and endometrium
- (D) Yolk sac and endometrium

41. Which of the following events occurring during the ovarian cycle is a result of the negative feedback caused by an increase in oestrogen levels?

- (A) Decreased production of progesterone at the end of the secretory phase
- (B) Increased secretion of luteinizing hormone during the follicular phase
- (C) Increased secretion of gonadotrophin-releasing hormone during the follicular phase
- () Decreased secretion of follicle-stimulating hormone after the release of the secondary oocyte

Item 42 refers to the following diagram which shows a longitudinal section through the ovule of an angiosperm.



42. The role of Y is to

- () fuse with a male nucleus to form the endosperm
- (B) fuse with a male nucleus to form the cotyledon
- (C) support the position of the embryo
- (D) support the position of the ovum

Items 43–44 refer to the following hormones.

- (A) Oestrogen
- (B) Progesterone
- (C) Luteinizing hormone
- (D) Follicle-stimulating hormone

43. Which hormone stimulates the proliferation of the endometrium?
44. Which hormone regulates the precise timing of ovulation?

45. The mini birth control pill works MAINLY by

- (A) suppressing follicle maturation and preventing ovulation
- (B) interfering with the penetration of the ovum by spermatozoa
- (C) preventing implantation of the fertilized ovum in the endometrium
- () thickening the cervical mucus thus preventing the spermatozoa from meeting the ovum